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ORIGINAL ARTICLE

Content and stability of B complex vitamins in commercial cosmetic products

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Abstract

Background: Individual B vitamins have many favorable effects on the skin and are common cosmetic ingredients. However, their formulation is demanding due to stability issues, which consequently affect the products' quality.

Aims: We aimed to determine the quality (labeling accuracy, content determination, and content-related quality control) and stability under long-term and accelerated storage conditions of a representative sample of commercial cosmetics containing the most common B vitamins – nicotinamide, dexpanthenol, pyridoxine, and cyanocobalamin.

Methods: Cyanocobalamin was determined by a previously published stability-indicating HPLC– diode array detector (DAD) method for the simultaneous determination of all hydrophilic vitamins. This method was additionally simplified and adjusted for the time-effective analysis of nicotinamide, dexpanthenol, and pyridoxine. Both methods were properly validated.

Results: All labeled B vitamins were present in the 36 tested products, mostly in contents, reported effective on the skin. Thus, a straightforward correlation between vitamin contents and product prices were not observed. The content-related quality control of eight products, which quantitively specify their content, revealed significantly lower nicotinamide contents (47% and 57%) in two products and appropriate or higher nicotinamide (102%–112%) and dexpanthenol (100%–104%) contents than declared in the remaining products. The 6-month long-term and accelerated stability studies demonstrated the products' physical stability, but also revealed dexpanthenol, pyridoxine, and cyanocobalamin degradation, while nicotinamide was mostly stable in the tested products.

Conclusions: The obtained results provide an inside into the quality of commercial vitamin B cosmetics and highlight the importance of stability testing in the formulation of quality, efficient, and safe cosmetics.

KEYWORDS
cosmeceuticals, cosmetic formulation, D-panthenol, niacinamide (nicotinamide), stability, vitamin B3

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1 | INTRODUCTION

B complex vitamins constitute a group of eight compounds including thiamin, riboflavin, niacin, pantothenic acid, pyridoxine, biotin, folate, and cobalamin, with several specific functions in the human body. All B complex vitamins have also found their place in the cosmetic industry. While some are mostly used as excipients (e.g., riboflavin as colorant), others are established (nicotinamide, dexpanthenol, and biotin) or upcoming (thiamin, pyridoxine, folic acid, and cyanocobalamin) topical agents in dermatology and cosmetology with valuable effects on the skin.

Thiamin and riboflavin are used in the treatment of rosacea and seborrheic diseases, due to the antipruritic effect of thiamine and the anti-inflammatory effect of riboflavin.[1] Because of the various beneficial effects on the skin, niacin in the form of nicotinamide is widely used as the (cosmetically) active ingredient in cosmetic and dermatological formulations. Nicotinamide has anti-microbial, anti-inflammatory, antipruritic, vasoactive, lightening, barrier- and photo-protective, and sebostatic effects on the skin.[2] Therefore, it is effective in the treatment of acne, atopic dermatitis, skin aging, melasma, light damage of the skin, hyperpigmentation, psoriasis, pellagra, nonmelanoma skin cancer, and other skin diseases and conditions.[2,3] Similarly, topical use of dexpanthenol, the alcohol analog of pantothenic acid, is widespread for the treatment and care of damaged skin, wound healing, atopic dermatitis, dermatitis, and scar management, due to its moisturizing, anti-inflammatory and effects on fibroblast proliferation.[4] Pyridoxine has been recently found to specifically increase filaggrin transcription and production in the skin, making it a candidate for the treatment of atopic dermatitis.[5] Biotin acts as a hair and skin-conditioning (miscellaneous) agent and is thus commonly found in hair conditioners and masks as well as skincare products.[1,6] Folic acid is an upcoming agent in the treatment of photodamaged and aged skin due to its involvement in cell replication, regulation of gene activity, and skin renewal.[7] The topical application of vitamin B12 is also a new approach in the treatment of atopic dermatitis and childhood eczema. Its beneficial effects are mediated through suppression of cytokine production, cell proliferation enhancement, and modulation of lymphocyte activity.[8,9]

Due to the many beneficial effects on the skin, B complex vitamins are common ingredients, added in different amounts in various cosmetic products, depending on their form and intended use. The main challenge associated with the formulation of B complex vitamins is assuring their stability. Most B complex vitamins are unstable and sensitive to various environmental factors (heat, oxygen, and light).[10,11] Their instability in cosmetic products may lead to their degradation, which affects the quality and also efficacy, and safety of the product. The instability of B complex vitamins may also be associated with the formation of degradation products with different and unwanted effects on the skin. Typical examples include the hydrolysis of niacinamide and nicotinate esters to nicotinic acid, which induces intense skin reddening even at low concentrations,[11] folic acid photodegradation to 6-formylpterin and pterin-6-carboxylic acid, which are associated with the generation of reactive oxygen species and skin photocarcinogenesis,[12] and the photodegradation of pyridoxine to phototoxic products, associated with skin photodamage and carcinogenesis.[13]

Cosmetic products on the EU market are not subject to regular quality control by the authorities, and their quality depends on the manufacturer. Cosmetics in the EU are regulated under Regulation EC 1223/2009, which states that their sampling and analysis should be carried out in a reproducible and standardized manner to ensure the uniform application and control of the restrictions for substances,[14] whereas control of the active ingredients content and stability evaluation are not required. Therefore, the growth of the cosmetic products market is accompanied by a growing concern regarding their quality by their users, some (accountable) manufacturers, and the scientific public. The latter suggests stricter regulation and quality control, following the principles of the pharmaceutical industry and using an appropriate analytical methodology.[15–17] While there are some publications on the quality control and stability evaluation of lipophilic vitamins in commercial cosmetic products,[15,16,18] such studies have not yet been reported for B complex vitamins. Considering their instability and its association with the quality, efficacy, and safety of cosmetic products, we aimed to determine the quality and stability of a representative sample of commercial leave-on cosmetic products containing the most common B complex vitamins—nicotinamide, dexpanthenol, pyridoxine, and cyanocobalamin. For such purposes, we used proper analytical methodology, based on conventional HPLC–DAD systems.

2 | MATERIALS AND METHODS

2.1 | Chemicals and reagents

Nicotinamide (B3) (≥99%) and pyridoxine chloride (B6) (≥99%) were purchased from Sigma-Aldrich. Dexpanthenol (B5) (≥98%) and cyanocobalamin (B12) (≥98%) were purchased from Biosynth. HPLC grade acetonitrile, methanol, and *n*-hexane were obtained from Honeywell, Riedel-de Haën™. Hydrochloric acid and sodium dihydrogen phosphate monohydrate were obtained from Merck. Ultrapure water was obtained through a Milli-Q water purification system A10 Advantage (Millipore Corporation).

2.2 | Instrumentation and chromatographic conditions

An Agilent Infinity 1290 ultra-high-performance liquid chromatograph (UHPLC) system (Agilent Technologies), equipped with a diode array detector (DAD) using a Max-Light highly sensitive cartridge cell (60 mm) and EZChrom acquisition system was used for the analysis. The examined vitamins were analyzed on a reversed-phase XSelect CSH C18 150 × 4.6 mm, 3.5 μm particle size column (Waters Corporation) at 40°C using 25 mM $NaH_2PO_4 \times H_2O$, pH adjusted to 4.0 (solvent A) and methanol (solvent B). The flow rate was 1 ml/min. Cyanocobalamin was analyzed according to a previously published

method for the simultaneous analysis of all main hydrophilic vitamins using gradient elution as follows: 0–4 min: 1% B, 4–8 min: 1%–16.5% B, 8–18 min: 16.5% B, 18–23 min: 16.5%–25% B, 23–25.2 min: 25% B, 25.2–26 min: 25%–1% B. The total run time was 30 min and the detection wavelength was 362 nm.[10] Nicotinamide, dexpanthenol, and pyridoxine were analyzed using isocratic elution (10% B) at 210 nm (dexpanthenol), 260 nm (nicotinamide), and 287 nm (pyridoxine) in a total run time of 8.5 min. Injection volume was adjusted to the individual cosmetic product and was between 1 and 10 μl.

2.3 | Preparation of standard solutions

B vitamins stock standard solutions were prepared fresh daily for three consecutive days of the validation by dissolving appropriate amounts of the individual vitamin in Milli-Q water. Six calibration standards (five in the case of cyanocobalamin) and three quality control (QC) solutions at low (QC_L), medium (QC_M), and high (QC_H) concentration levels (in triplicate) (Table 1) were prepared by diluting the individual stock standard solution with Milli-Q water.

2.4 | Method validation

The previously published stability-indicating HPLC–DAD method used for the cyanocobalamin analysis as well as the simplified isocratic HPLC–DAD method for the simultaneous quantification of nicotinamide, dexpanthenol, and pyridoxine were validated following the ICH guidelines Q2(R1)[19] in terms of specificity, linearity, precision, accuracy, detection limit (LOD), quantitation limit (LOQ), and stability of the QC solutions. Within validation of the analytical procedure, recovery, and sample preparation repeatability were evaluated.

To evaluate the specificity of the methods, we evaluated chromatograms of the individual vitamin standard solutions, their mixtures, the used solvents (methanol and Milli-Q water), and also all the tested cosmetic products for potential interferences.

TABLE 1 Nicotinamide (B3), dexpanthenol (B5), pyridoxine (B6), and cyanocobalamin (B12) concentrations in the calibration standards and quality control (QC) solutions

		Concentration [mg/L]			
		B3	B5	B6	B12
Calibration standards		7.50	4.69	0.75	0.04
		18.8	9.38	1.88	0.09
		37.5	18.8	3.75	0.19
		75.0	37.5	7.50	0.30
		150	75.0	15.0	0.38
		300	150	30.0	/
QC solutions	QC_L	22.5	11.3	2.25	0.11
	QC_M	67.5	33.8	6.75	0.23
	QC_H	225	113	22.5	0.34

The linearity of the methods was evaluated using six calibration standards (five in the case of cyanocobalamin) (Section 2.3). The method of least squares was applied in regression analysis to determine the calibration line equations and the correlation coefficients (R^2). The acceptance criterion was $R^2 > 0.999$.

The accuracy and precision of the methods were determined based on the QC solutions on three concentration levels, covering the entire linear range (low, medium, and high), prepared in triplicate each day of the validation (Section 2.3). Accuracy was determined as a ratio (%) of the determined and actual concentration. The acceptance criterion was 100 ± 5%. Precision was determined as a relative standard deviation (RSD) of the responses of the three parallels of each QC solution and was expected to be ≤5%. Injection repeatability was also evaluated by six consecutive injections of one parallel of each QC solution (QC_L, QC_M, and QC_H) and was expected to be ≤2%.

LOD and LOQ were calculated from the calibration lines, using the following equations: $LOQ = (10 \times \sigma)/S$; $LOD = (3.3 \times \sigma)/S$ where σ is the SD of the intercepts, and S is the average slope of the calibration curve. The calculated LOQ and LOD values were confirmed by the analysis of standard solutions with approximately the same concentrations.

The stability of the QC solutions was evaluated on each QC level within 48 hours of storage at 25°C. Sample stability was expressed as a percentage of the response after 24 and 48 hours in relation to the response at time 0 and was expected to be within 100 ± 10%.

Method recovery was determined individually for particular vitamins: nicotinamide, dexpanthenol, and cyanocobalamin. A cosmetic product with a significant amount of the evaluated vitamin was selected and analyzed with and without the addition of approximately the same vitamin amount as contained in the product. Standard solutions of the added vitamin amount were also analyzed. All the samples were prepared in triplicate. Average recoveries were calculated using the following equation: recovery (%) = 100 × (concentration found in spiked sample − concentration found in the non-spiked sample)/added concentration. The acceptance criterion was 100 ± 10%.

Sample preparation repeatability was determined as the RSD of the responses of each cosmetic product, prepared and analyzed in triplicate. The acceptance criterion was 5%.

2.5 | Tested commercial cosmetic products

A total of 33 cosmetic products, 2 creams, registered as medical devices, and one registered as a medicinal product were tested. All the products were obtained on the Slovenian market in May 2021 and included serums (Balea, La Roche-Posay, Mabox, Missha, Paula's choice, Revolution, Revox, The Ordinary), face creams (Afrodita, Cien (two products), Darphin Paris (two products), Ducray Laboratories Dermatologiques, Eubos, Freezyderm, La Roche-Posay (two products), Paula's choice, Rilastil, Skintegra (two products)), eye cream (La Roche-Posay), tonic (Afrodita, Balea, Geek & Gorgeous), balms (La Roche-Posay [three products]), gels (Afrodita, La Roche-Posay), lotion (Revox), face and

body creams (Bayer (two products), Dr. Derm, Eucerin). The tested cosmetic products belonged to different price ranges (1–120€ calculated per 50 ml of the cosmetic product).

2.6 | Commercial cosmetic products sample preparation

We developed the sample preparation procedure based on our experience and previously published procedures for the extraction of lipophilic vitamins.[15,16] The optimal extraction solvent, its volume, and the amount of the weighted cosmetic product in the sample preparation procedure were determined within preliminary testing and were adjusted to each product. Samples from the obtained cosmetic products were prepared in triplicate and analyzed immediately after their opening.

A predefined amount of the cosmetic products (25–200 mg) was initially weighted into a centrifuge tube, followed by the addition of 5–10 ml of Milli-Q water (in the case of liquid cosmetic products) or 10 ml of methanol or Milli-Q water (in the case of semi-solid cosmetic products). The samples were further vortexed (3 min), sonicated (15 min), vortexed once again (2 min), and centrifuged (16 200 × *g*, 25°C, 10 min). If needed, the samples were also filtered through an RC filter with a 0.45 μm pore size and transferred into vials for analysis.

2.7 | Stability study

The stability of selected cosmetic products was evaluated under accelerated and long-term storage conditions in climate chambers, following the CTFA (Cosmetic, Toiletry, and Fragrance Association) and COLIPA (European Cosmetic Toiletry and Perfumery Association—now Cosmetic Europe) guidelines[20] as well as the ICH Q1A (R2) guidelines.[21] Long-term testing was performed on cosmetic products stored at 25°C and 60% relative humidity (RH) in their original packaging. Accelerated testing was performed by exposing previously weighed samples in centrifuge tubes to elevated temperature and RH (40°C and 75% RH). Changes in the tested vitamin contents and organoleptic properties of the cosmetic products were evaluated at different time points for 6 months: after 1 week, 2 weeks, 3 weeks, 1, and 6 months. Samples were prepared in triplicate at each time point following the sample preparation procedure, described in Section 2.6.

3 | RESULTS AND DISCUSSION

3.1 | Selection of analytes, commercial products, and analytical methods

The main aim of this study was to evaluate the content and stability of frequently present B complex vitamins in commercial cosmetic products and thus determine their quality. Therefore, we focused on leave-on cosmetic products, whose quality interrelates with their efficacy and safety due to their prolonged contact with the skin. Based on the performed screening of available leave-on cosmetic products on the Slovenian market, we concluded that nicotinamide and dexpanthenol are the main representatives of B complex vitamins in commercial cosmetics. Pyridoxine and cyanocobalamin are less often found, whereas other B complex vitamins (biotin and folic acid) are very rarely found in commercial cosmetics on the Slovenian market. Therefore, such vitamin distribution is also maintained within our range of tested commercial cosmetic products, which included products in various price ranges and formulations in liquid and semi-solid forms from different manufacturers and also different products from the same manufacturers. In view of the quality control of cosmetic products in relation to their declarations and labeling, products with declared vitamin content were also included in the study. For comparison purposes, representatives of B complex vitamins in similar formulations but in more strictly regulated products—medical devices and medicinal products were also included. The basis for content and stability evaluation is proper analytical methodology. A previously published stability-indicating HPLC–DAD method for the simultaneous determination of all main hydrophilic vitamins[10] was used for the evaluation of cyanocobalamin. This method was additionally simplified and shortened resulting in an isocratic HPLC–DAD method for the simultaneous quantification of nicotinamide, dexpanthenol, and pyridoxine. The sample preparation procedure was developed and optimized for the individual cosmetic products.

3.2 | Validation of the analytical methods

Both used HPLC–DAD methods were validated following the ICH guidelines Q2(R1) in terms of specificity, linearity, precision, injection repeatability, accuracy, LOD, LOQ, and stability of QC solutions.[19] Both methods were confirmed specific as no interferences were detected in the chromatograms of the used solvents, vitamins standard solutions and their mixtures, and the tested products. Representative chromatographs of cyanocobalamin standard solution and a standard mixture of nicotinamide, dexpanthenol, and pyridoxine are presented in Supplementary materials (Figures S1 and S2). The remaining validation data are presented in Table 2. The selected concentration ranges cover the evaluated vitamins' concentrations in the analyzed products' samples. The determined high R^2 demonstrates the linearity of the methods. The sensitivity of both methods for the analysis of the evaluated vitamins in commercial products was satisfactory. LOQ was <2.00 mg/L, which corresponds to up to <0.001% (w/w) assuming maximum limit values in the sample preparation parameters (5 ml of extraction solvent, 200 mg of the cosmetic product, and 10 μl injection volume) in the case of nicotinamide and dexpanthenol and significantly lower (approximately 4-fold) for pyridoxine and (25-fold) for cyanocobalamin (Table 2). Moreover, both methods were also found accurate and precise, as the obtained results for accuracy, precision, and injection

TABLE 2 Validation data of analytical methods

	Range [mg/L]	LOD [mg/L]	LOQ [mg/L]	R^2	Accuracy [%] (AV ± SD)	Precision [%]	Injection repeatability [%]	Stability[a] [%] (AV ± SD)
B3	7.50–300	0.66	2.00	1.0000	100.1 ± 0.5	0.3	0.2	100.1 ± 1.8
B5	4.69–150	0.63	1.88	0.9999	100.4 ± 0.7	0.7	0.8	100.6 ± 0.9
B6	0.75–30.0	0.15	0.46	1.0000	100.4 ± 0.5	0.3	0.2	100.9 ± 4.9
B12	0.09–0.38	0.03	0.08	0.9997	104.5 ± 2.4	2.1	1.0	97.4 ± 1.5

[a]after 48 h of storage at 25°C.

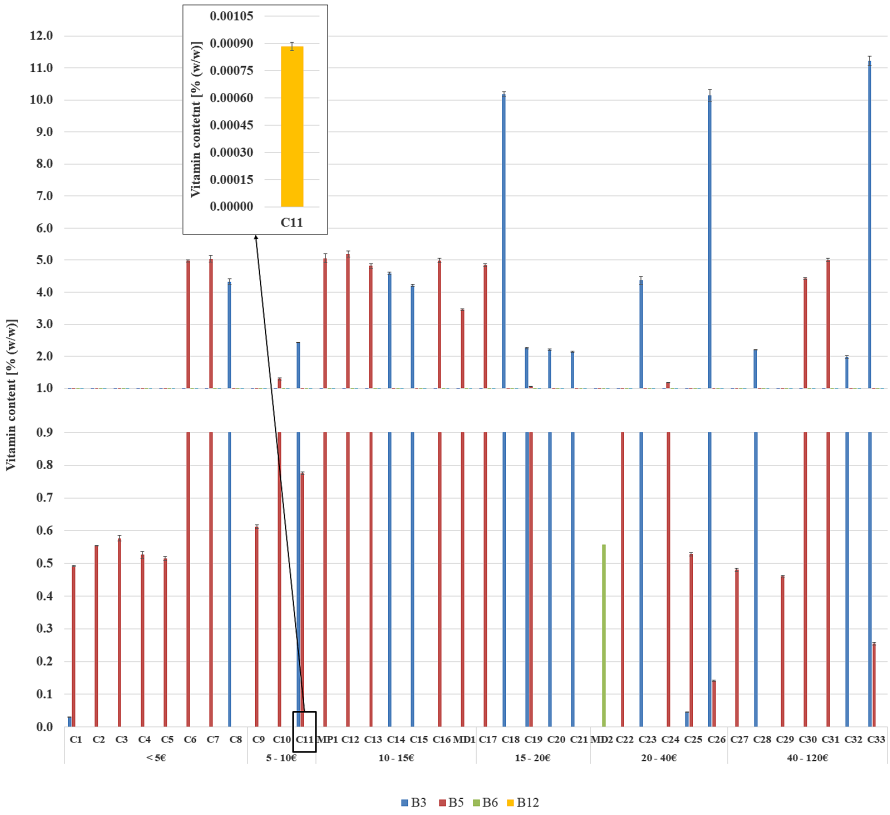

FIGURE 1 Determined contents of nicotinamide (B3), dexpanthenol (B5), pyridoxine (B6), and cyanocobalamin (B12) in the tested cosmetic products (C), medical devices (MD), and medicinal product (MP), expressed as a mass percentage (%) (average ± SD, [$n = 3$]) in relation to the price of the products (per 50 ml). *the labels of individual products are retained throughout the manuscript.

repeatability are within the acceptance criteria. The repeatability of the sample preparation procedure was demonstrated for all tested products (RSD < 5% between the triplicates), as displayed in Figures 1 and 2. The average recoveries were between 95.2% and 104.5% and were also within the acceptance criteria. The evaluated vitamins were also sufficiently stable (mostly within 100 ± 5%) after 48 h of storage at 25°C.

3.3 | B complex vitamins content in the tested commercial products

Within the performed quality control, we initially evaluated whether the labeled B complex vitamins are present in the tested products. Thus, we confirmed the proper labeling of B complex vitamins in all tested products, as the labeled vitamins were present in the stated form and additional, not labeled B complex vitamins were not observed. In the tested range of commercial products, dexpanthenol was the most prevalent and was present in 26 products, followed by nicotinamide (15 products) and pyridoxine and cyanocobalamin—each present in one product. The results of their quantitative evaluation in the tested products are summarized in Figure 1.

The determined dexpanthenol contents in the 24 tested cosmetic products ranged between 0.141% and 5.191% (w/w) in the tested products and were similar in the different tested product groups: between 0.141% and 5.053% (w/w) in the tested creams, 0.254% and 5.191% (w/w) in the tested serums, and between 0.492% and 4.980% (w/w) in the tested tonics. Such ranges are somewhat expected in view of the efficacy of the products and are also within the range of used dexpanthenol concentrations in >5,000 cosmetics, reported by their manufacturers within the Food and Drug Administration's Voluntary Cosmetic Registration Program (VCRP).[22] Within interpretation of the results, we observed that dexpanthenol contents in the tested cosmetic products tend to gravitate towards two typical values – around 0.5% (w/w) and around 5% (w/w). Namely, almost a third (30%) of the tested cosmetic products contained around 0.5% (0.45%–0.55%) (w/w) dexpanthenol and another third (33%) of the tested cosmetic products contained around 5% (4.5%–5.5%) (w/w)

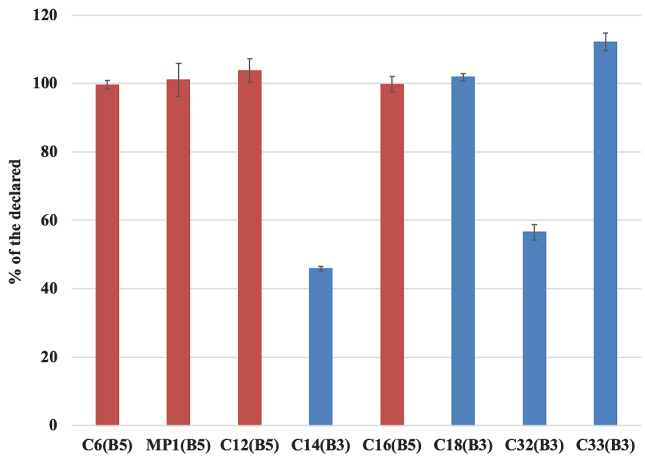

FIGURE 2 Determined contents of nicotinamide (B3) and dexpanthenol (B5) in the tested medicinal (MP) and cosmetic products (C) with their declared contents, expressed as an average % of the declared content (average ± SD, [$n = 3$]).

dexpanthenol. According to literature data, dexpanthenol in concentrations between 2% and 5% (w/w) is effective in the treatment of wounded or irritated skin,[4] which is the range of its determined content in the tested medicinal product and medical device (Figure 1). Formulations with 1% (w/w) of dexpanthenol have a moisturizing effect on the skin, whereas lower dexpanthenol concentrations (0.5% (w/w)) do not have a significant effect on skin barrier and skin moisture.[23] Based on the obtained results, we conclude that half of the tested cosmetic products contain dexpanthenol in concentrations, which are effective on the skin according to literature data (≥1% (w/w)). These products are scattered among all price ranges and are most concentrated among the affordable cosmetics (10–15€ per 50 ml of the cosmetic product, Figure 1). The remaining 12 cosmetic products contained between 0.141% and 0.775% (w/w) dexpanthenol, which is significantly below its effective concentration of 1% (w/w). These products were most common among the lower-priced cosmetics (C1-5 in Figure 1). However, considering their occurrence in each price range, also including three (C27, C29, and C33) of the five tested cosmetic products in the highest price range, we conclude that higher price of the product is not necessarily related to higher dexpanthenol content (Figure 1).

The determined contents of nicotinamide in the 15 tested cosmetic products ranged between 0.030% and 11.217% (w/w). Similarly, as in the case of dexpanthenol, comparable ranges of nicotinamide contents were determined in different product forms (e.g., serums, creams). Thereby, the determined nicotinamide contents gravitated toward three typical concentrations: 2.0%–2.5% (w/w) (40% of the tested products), 4.2%–4.6% (w/w) (27% of the tested products) and 10.1%–11.2% (w/w) (20% of the tested products). The majority (13) of the 15 tested cosmetic products contained nicotinamide in concentrations around or above the ones reported in the literature to improve skin appearance. Namely, the topical application of nicotinamide in concentrations between 2% and 5% (w/w) has been shown effective in the treatment of acne, and atopic dermatitis, and as a lightening, anti-aging, and anti-psoriatic agent.[3,24] About 20% of the tested products contained high-nicotinamide concentrations (around 10% (w/w)), whose effects have not been investigated in the literature. However, clinical testing of nicotinamide in concentrations up to 10% (w/w) showed no stinging sensation.[25] Also, as nicotinamide is not a sensitizer or photosensitizer, it is considered safe in the current concentrations and practices of use, and its concentration is therefore not restricted in cosmetic products in the EU.[25] Evaluating the relationship between nicotinamide content and the prices of the products, we did not observe a straightforward correlation (e.g., an increase in nicotinamide content with an increase in the products price). Although all products with high-nicotinamide content (>5% (w/w)) belonged to the mid or higher price range (15–120€ per 50 ml of the cosmetic product), these price ranges also comprise products with some of the lowest determined nicotinamide contents (C24 and C32 in Figure 1).

The use of pyridoxine and cyanocobalamin in topical (cosmetic) products is upcoming and thus their effective concentrations have not yet been established. Also, they are quite rare ingredients in cosmetics products and their contents have not yet been reported. The determined pyridoxine content in the one tested product was 0.557% (w/w). The determined cyanocobalamin content in the one tested cosmetic product was significantly lower (8.8×10^{-4}% (w/w)).

3.4 | B complex vitamins content-related quality control of the tested commercial products

Eight of the 36 evaluated commercial products (one medicinal product and seven cosmetic products) quantitatively specified the content of nicotinamide (four products) and dexpanthenol (four products). With the increasing number of such products on the market and their insufficient control by the official authorities, there is also a growing need for their proper quality control, which was performed within this study. The content-related quality control was assessed by comparing the determined and the declared content (Figure 2).

As can be seen in Figure 2, all four tested cosmetic products contained the declared dexpanthenol amount (100 ± 5% of the declared content). Contrarily, the determined nicotinamide content was as stated (101.8% of the declared content) in one tested cosmetic product, higher than declared (112.2%) in one cosmetic product, and significantly lower (46.9% and 56.5%) in the remaining two cosmetic products. Following the principles on acceptable tolerances for vitamins content in foods and food supplements, established by the European Commission based on their safety, on one hand, and general instability on the other,[26] we consider the determined deviations between the contained and declared vitamin amount in most products (C6, C12, C16, C18, C33, and MP1, Figure 2) acceptable. Contrary, as nicotinamide content significantly deviate from the labeled one in C14 and C32 (Figure 2), these two cosmetic products are inappropriate in view of their content-related quality. Although this is the first study, disclosing inappropriate vitamin B contents in relation to the declared contents in cosmetic products, such findings have been previously reported in food supplements.[10,27] Possible

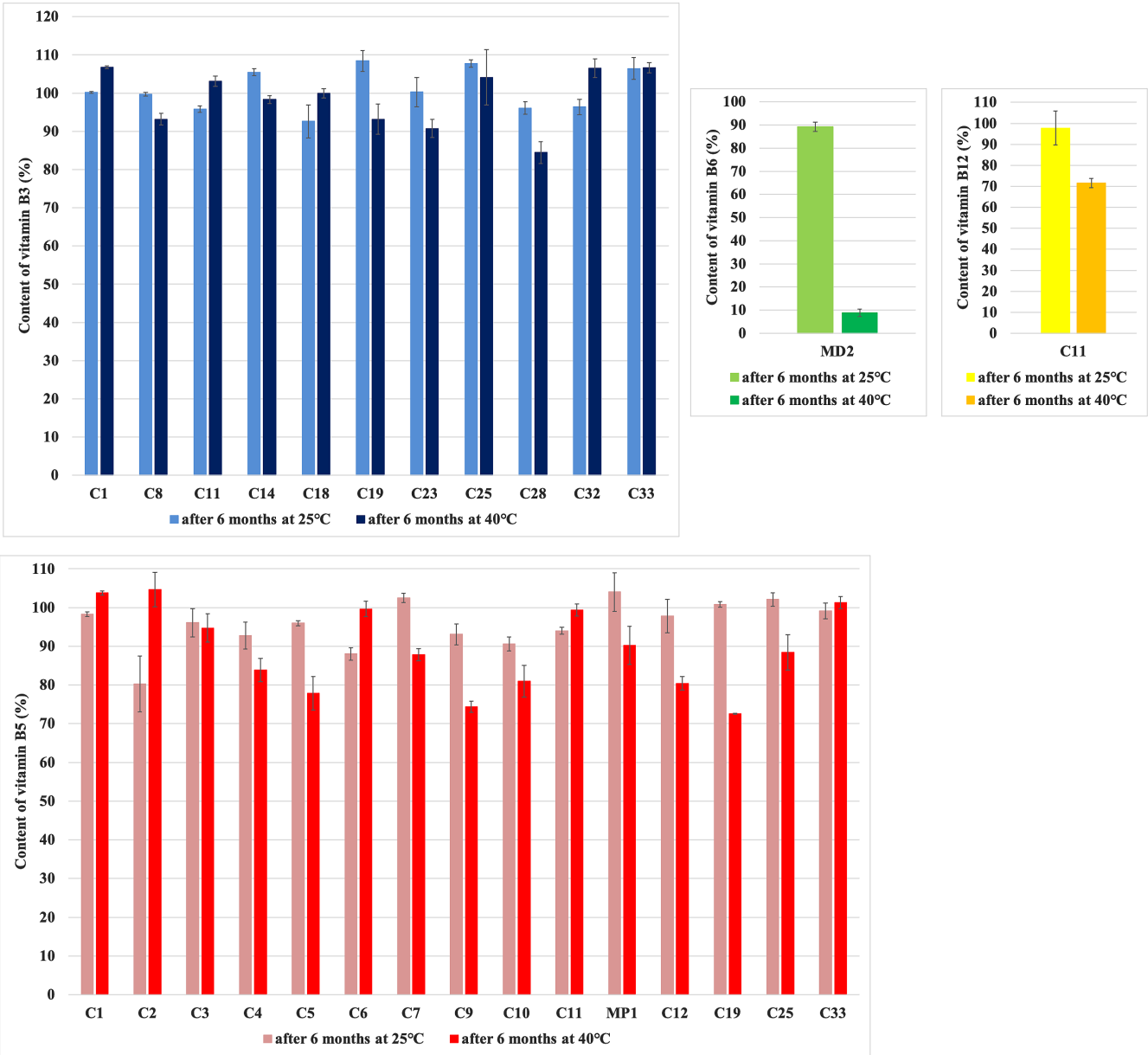

FIGURE 3 The remaining content of nicotinamide (B3), dexpanthenol (B5), pyridoxine (B6), and cyanocobalamin (B12) in the tested cosmetic products (C), medical devices (MD), and medicinal product (MP) after 6 months of storage at 25°C and 40°C, expressed as a % of the initial content (average ± SD, [$n = 3$]).

reasons for their inappropriate content include manufacturing inaccuracies, use of ingredients of insufficient quality, intentional addition of higher or lower amounts than declared, or inappropriate formulation in view of the stability and compatibility, which are all associated with the lack or inappropriate quality control.[10,15,27]

3.5 | B complex vitamins stability in the tested commercial products

To assess the overall stability of the tested commercial products and evaluate whether stability issues are a reason for the determined lower nicotinamide content in relation to the declared content in two cosmetic products, a stability study was performed following the CTFA–COLIPA and ICH Q1A(R2) guidelines.[20,21] The stability of B complex vitamins was evaluated in a total of 22 products, including all products with quantitatively specified contents, except C16. The physical and chemical stability of the products was evaluated at regular time points during 6 months of storage at long-term (25°C and 60% RH) and accelerated stability testing conditions (40°C and 75% RH). No changes in the organoleptic properties were observed during the long-term and accelerated stability study, thus confirming the physical stability of all tested commercial products. Evaluating the chemical stability, we detected certain changes, which were specific to the individual vitamins and formulations (Figure 3). Based on the obtained results,

we may confirm the stability of nicotinamide in the 11 evaluated products, as its content remained >92% of the initial content after 6 months of storage at long-term conditions and > 85% at accelerated storage conditions. Considering its high stability (>96% of the initial content) in the two cosmetic products (C14 in C32) with significantly lower nicotinamide content than declared (Figure 2), we conclude that the inappropriate quality of these products is not a consequence of its instability. Dexpanthenol was found less stable than nicotinamide, yet its instability was not directly correlated with the formulation type (liquid and semi-solid formulation). Significant dexpanthenol degradation (>10% of the initial content) was observed in two cosmetic products, stored at ambient temperature, and eight products stored at elevated temperature, in which up to 28% degradation was observed (Figure 3). Elevated storage temperature had a more pronounced effect on cyanocobalamin, which was stable when stored at ambient temperature (98% of the initial content after 6 months) and significantly degraded (28% degradation) when stored at elevated temperature and especially on pyridoxine (10% degradation at ambient temperature and > 90% degradation at elevated temperature) (Figure 3). The stability of the most common B complex vitamins in commercially available cosmetic products has not yet been described in the literature. However, the obtained results are supported by several reports on their stability in different samples (solutions, foods, and pharmaceutical formulations) in the literature, where nicotinamide is ascertained as quite stable under various (stress) conditions, and dexpanthenol, pyridoxine, and cyanocobalamin as differently susceptible to degradation in different samples and conditions.[10,28,29] The observed higher degradation at elevated storage temperature is in line with the general principles and somewhat expected, as the temperature is one of the key environmental factors, affecting vitamin stability in general, although, contradictory information regarding its effect on the stability of dexpanthenol, pyridoxine, and cyanocobalamin may be found in the literature.[10,28,29] Considering the inconsistent data on vitamins stability in the literature and their susceptibility to the effects of various factors, including also the formulation itself, stability studies should be applied as a useful tool in the formulation of quality cosmetic products. Especially accelerated stability testing is gaining ground in the cosmetic industry, as it may enable the prediction of the product's stability under the desired storage conditions in a shorter time and provides information on the effect of short-term deviations from the expected storage conditions.[21] However, prediction of the stability of products containing active ingredients and other excipients based on accelerated stability testing is quite demanding and has reported limited applicability to certain cosmetic products.[18] Therefore, the results of accelerated stability testing should be confirmed by long-term testing, as done in the pharmaceutical industry, and should include both determination of the physical and chemical stability of the products. This is also demonstrated by the obtained results within this study, where all products remained physically stable, whereas temperature had very different effects on the chemical stability of individual vitamins. Namely, temperature significantly affected B complex vitamins stability in some products (MD2, C9, C11, and C19, Figure 3), but did not have the (expected) effect on other products (C1, C11, and C33, Figure 3).

4 | CONCLUSIONS

B complex vitamins have many favorable effects on the skin. Therefore, the topical application of nicotinamide and dexpanthenol is established in numerous skin conditions, while other B complex vitamins (e.g., pyridoxine and cyanocobalamin) are forthcoming agents in cosmetology and dermatology. In view of their widespread use in the cosmetic industry, which is associated with some formulation challenges and the absence of regular quality control by the authorities, we focused on the evaluation of the quality and stability of a representative sample of commercially available products with B complex vitamins. We thus evaluated the labeling accuracy, the content, and stability of the most prevalent B complex vitamins on the Slovenian cosmetic market—nicotinamide, dexpanthenol, pyridoxine, and cyanocobalamin in a total of 36 products. For such purpose, we used a previously developed HPLC–DAD method, which was additionally simplified and shortened. The performed quality control confirmed the presence of all labeled B complex vitamins in the tested products but also revealed inappropriate contents in two of the eight tested cosmetics with declared vitamin B content. All the tested products were found physically stable after 6 months of storage at long-term and accelerated storage conditions. Nicotinamide was found stable in the tested products, while the stability of dexpanthenol, pyridoxine, and cyanocobalamin depended on the formulation and the storage temperature. Even though stability is a fundamental aspect in the formulation of quality cosmetic products, within this study we revealed significant dexpanthenol, pyridoxine, and cyanocobalamin degradation under both long-term and accelerated stability conditions. Thus, with this article, we suggest a principle for quality control and stability testing for both manufacturers and regulatory authorities and make an effort towards better recognition of the importance of stability for the production of quality and safe cosmetics.

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ETHICAL APPROVAL

No ethical approval was required for this article.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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